

Techniques of Artificial Intelligence Breast Cancer Project

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May 2026

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0.1 Introduction

In the following paragraphs, we will cover basic information about the project outline, the dataset we have been working on and we will define the main research aims of the project.

0.1.1 Project Background

In 2022, there were an estimated 2.3 million women diagnosed with breast cancer and 670 000 deaths globally. The biological composition of the breast is different from other organs. Healthy tissue is similar in composition to fatty and cancerous tissue. There is a strict procedure for screening examinations. Those examinations include mammography. During mammography, there are strict guidelines about the dose that can be delivered to tissue, there for srceening examiantions most outperfrom risk multiple times. Until recent years, doctors would look at scans to see any abnormalities. In recent years, machine learning algorithms have found their use in mammography, detecting and assessing microcalcifications which are not visible to the eye. As a result, many different algorithms and approaches have been tested in previous years, and results are already seen and applied in clinical use, which makes it even more relevant.

0.1.2 Literature overview

1

Several studies have used radiomics and machine learning for breast cancer classification and recurrence prediction. A common challenge in these studies is the large number of extracted features. Therefore, feature selection is often used to remove redundant features, reduce overfitting, and improve model performance.

Granzier et al. [4] studied the robustness of MRI-based radiomic features in breast cancer. They compared features extracted using RadiomiX and PyRadiomics and assessed how stable these features were under inter-observer segmentation differences. The study showed that not all radiomic features are reproducible. This highlights the importance of selecting features that are reliable before using them in classification models.

Oladimeji et al. [5] extracted 857 radiomic features from digital breast tomosynthesis images using PyRadiomics. These features included first-order statistics and texture features such as GLCM, GLRLM, GLSZM, NGTDM, and GLDM. To reduce the feature set, the authors used Mutual Information-based Radiomic Feature Selection. This method selected the most informative features and reduced the effect of redundant variables. Most selected features came from wavelet-transformed images. The selected features were then tested using classifiers such as Logistic Regression, Decision Tree, Random Forest, and XGBoost.

¹This section summarises related work on radiomics and machine learning for breast cancer prediction. Text was generated via ChatGPT 5-Thinking model. However, AI was not used to find references or cite anything. References were searched on Google Scholar using terms such as breast cancer classification, pyradiomics, radiomics feature extraction, evaluation of ML metrics for breast cancer, breast cancer classification deep learning and the Wisconsin Breast Cancer Dataset. Full prompt and text provided to AI can be found at this link [ChatGPT prompt](#)

Al Reshan et al. [2] used Recursive Feature Elimination with cross-validation to select important features for breast cancer classification. This helped reduce dimensionality and limit overfitting. The study also tested ensemble learning methods, including bagging, voting, boosting, and stacking. These methods combine several models to improve prediction. The base classifiers included Logistic Regression, Support Vector Classifier, K-Nearest Neighbours, XGBoost, Gaussian Naive Bayes, Decision Tree, and Random Forest.

Agarap [1] compared several machine learning models on the Wisconsin Diagnostic Breast Cancer dataset. The dataset includes features that describe cell nuclei, such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension. The study tested models including GRU-SVM, Linear Regression, Multilayer Perceptron, Nearest Neighbour search, Softmax Regression, and Support Vector Machine. The Multilayer Perceptron achieved the best reported performance.

Fan et al. [3] used ultrasound radiomics to predict breast cancer recurrence risk. The authors extracted features with PyRadiomics and selected the final features using mRMR and LASSO. The final radiomics signature included 12 features. One of the most important features was wavelet-LHL first-order Mean. The study showed that combining clinical, ultrasound, and radiomic information improved recurrence prediction.

Overall, previous studies show that feature selection is an important step in breast cancer machine learning tasks. It helps reduce unnecessary features, improves model stability, and can support better classification performance. These studies also show that wavelet-based radiomic features and ensemble learning methods are commonly useful in breast cancer prediction models.

0.1.3 Aim and Objectives

0.2 Dataset and Exploratory Data Analysis

0.2.1 Dataset description

² The dataset we used in the project contained a total of 96 female patients who underwent a biopsy at UZ Brussels, after an earlier mammography examination indicated the presence of suspicious MCs. From each patient, a tissue sample of 3 mm diameter and a length of 23 mm was extracted and embedded in paraffin. The final benign/malignant diagnosis is based on histopathological analysis of these samples. A sample is considered malignant if cancer cells are detected. Benign MCs could thus be present in malignant samples and, therefore, be labelled.

0.2.2 Exploratory Data Analysis

The first step was to examine the dataset and assess its overall quality. The dataset contained a total of 3,562 instances from 96 patients, with 151 feature columns and one final column named Label, which indicated the class using binary values of 0 and 1. Before selecting features, the data quality was checked because features with a

²Taken from them project description

large proportion of missing values could negatively affect model performance. Therefore, the dataset shape, column names, data types, missing values, and duplicate entries were examined. No abnormalities were detected, and the data appeared to be consistent across the feature columns. However, class imbalance was observed, with 2,020 benign instances and 1,542 malignant instances. The distribution of instances across patients was also examined to determine whether each patient contributed a similar number of instances. This showed that the distribution was not balanced and was right-skewed, as shown in the graph below. The mean number of instances per patient was 37.1 ± 44.3 , while the median was 25.5. Therefore, we decided to account for this imbalance during model development.

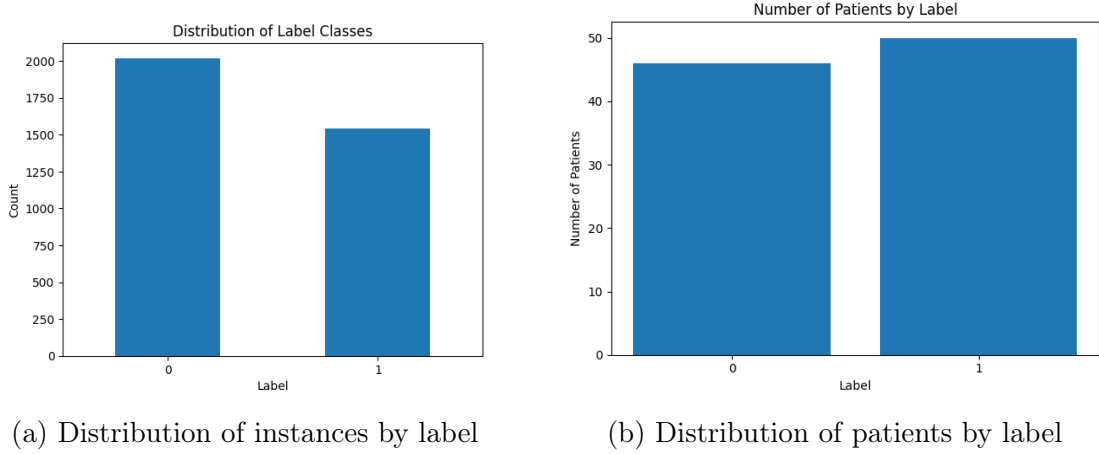


Figure 1: Class distribution at instance and patient level

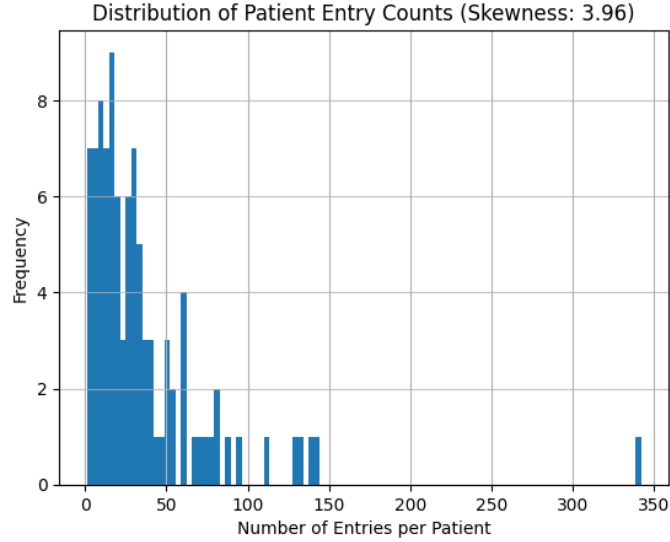


Figure 2: Distribution of skewness values

0.2.3 Feature Extraction

The dataset contained a total of 151 feature columns. Therefore, the extracted features were examined before model development. Most of the features were wavelet-

based, with 121 columns containing the term wavelet. In addition, 18 columns contained the term original, and 11 columns contained the term log. The original features were extracted directly from the original image data and included first-order features, such as kurtosis and skewness, as well as 3D shape features, such as elongation and spherical disproportion.

Several texture feature groups were also present in the dataset. Grey Level Dependence Matrix (GLDM) features describe how much a voxel depends on neighbouring voxels with similar intensity. These features are useful for measuring local homogeneity, roughness, intensity variation, and repeated texture structures. Grey Level Run Length Matrix (GLRLM) features describe consecutive runs of voxels with the same or similar grey intensity in a specific direction. These features are useful for identifying fine or coarse texture, streak-like patterns, uniform regions, and directional texture. Grey Level Size Zone Matrix (GLSZM) features describe connected zones of voxels with the same or similar grey intensity. Unlike GLRLM, GLSZM features are not direction-dependent, and they are useful for analysing homogeneous region size, texture granularity, small versus large intensity zones, and tissue heterogeneity.

Some features were extracted after applying a Laplacian of Gaussian filter. In this dataset, features labelled as log-sigma-3-0-mm-3D were not extracted directly from the original image, but after applying a 3D Laplacian of Gaussian filter with $\sigma = 3.0$ mm. This filter highlights structures and intensity changes at that spatial scale.

Wavelet features were also included. These features were extracted after applying high-pass and low-pass filters in three dimensions. Subgroups such as LLL, LLH, and HHH indicate different combinations of low-pass (L) and high-pass (H) filtering. For example, LLL represents low-pass filtering in all three dimensions, while HHH represents high-pass filtering in all three dimensions. Features with a wavelet2 prefix indicate a second-level wavelet decomposition, which can capture additional fine or coarse image details. Information about these radiomic feature definitions was based on PyRadiomics [6].

0.2.4 Dataset Splitting

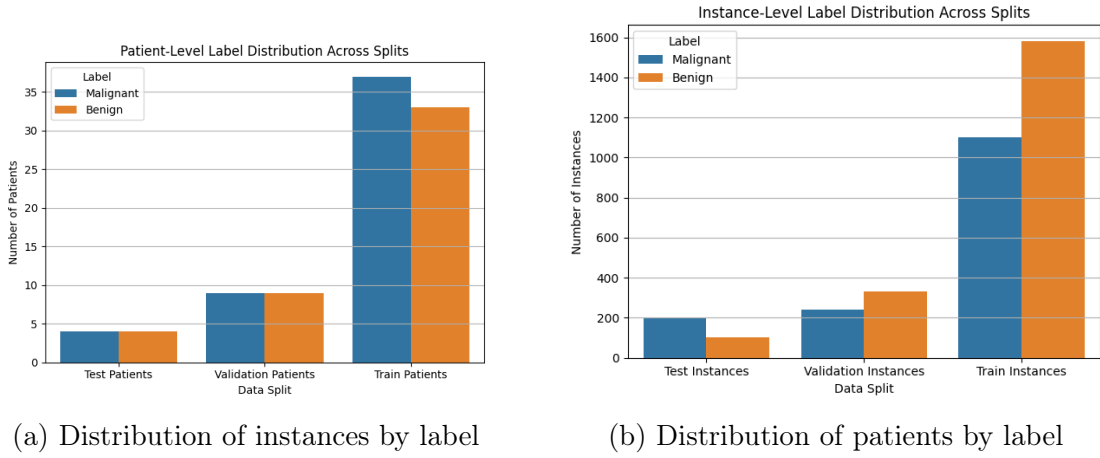
The dataset was split into training, validation, and test sets at the patient level. Although the test set was relatively small, it was kept separate so that the final selected classifier could be evaluated on unseen patients. The training set was used for feature selection and model fitting, the validation set was used to compare and select the best classifier, and the test set was reserved for final evaluation. Overall, the dataset contained 3,562 instances from 96 patients, with 46 benign patients and 50 malignant patients. In the Table you can see data after patient-level split and on the instance-based split. This approach guarantees that no patient’s instances appear across multiple splits, fully eliminating the risk of patient-level leakage.

Table 1: Patient-level dataset split

Dataset split	Purpose	Patients	Benign	Malignant
Train	Feature selection and model fitting	70	33	37
Validation	Classifier comparison and selection	18	9	9
Test	Final evaluation of best classifier	8	4	4
Total	Complete dataset	96	46	50

Table 2: Instance-level distribution across dataset splits

Dataset split	Instances	Patients	Benign instances	Malignant instances
Train	2686	70	1583	1103
Validation	576	18	333	243
Test	300	8	104	196
Total	3562	96	2020	1542



(a) Distribution of instances by label

(b) Distribution of patients by label

Figure 3: Class distribution at instance and patient level

0.3 Algorithms description

3

Logistic Regression is a linear classification algorithm. It estimates the probability that a microcalcification belongs to a benign or malignant class using a weighted combination of the input features. It is useful as a baseline model because it is simple, interpretable, and tests whether the classes can be separated using a mostly linear decision boundary.

Random Forest is a tree-based ensemble algorithm. It builds multiple decision trees using different subsets of the data and features. Each tree gives a prediction, and the final class is selected by majority voting. This method can capture non-linear relationships and interactions between radiomic features, which makes it suitable for complex microcalcification patterns.

K-Nearest Neighbours is an instance-based learning algorithm. It does not build an explicit model during training. Instead, it classifies a new microcalcification

³This text summarises machine learning algorithms that were used for our project. Following the last prompt, where AI was asked to reformulate text provided from articles or research papers, AI was asked to write a short description of each ML model itself. Prompt can be found at this link [ChatGPT prompt](#)

based on the labels of the closest training samples. It is useful for checking whether microcalcifications with similar radiomic feature values tend to belong to the same class.

Linear Discriminant Analysis is a statistical linear classification algorithm. It finds a linear combination of features that best separates the classes. It assumes that the feature values are normally distributed and that the classes have similar covariance. It provides an interpretable linear method for distinguishing benign and malignant microcalcifications.

Multilayer Perceptron is a neural-network-based algorithm. It contains input, hidden, and output layers. The hidden layers learn non-linear combinations of the input features through weighted connections and activation functions. It is used to test whether more complex learned relationships between features can improve classification.

Gaussian Naive Bayes is a probabilistic classification algorithm. It uses Bayes' theorem to estimate the probability of each class based on the feature values. It assumes that features are independent and normally distributed. Although this assumption may be simplified for radiomic data, it provides a fast and simple probability-based comparison model.

0.4 Methodology

0.4.1 Machine Learning Pipeline

In the following paragraph, we will explain in detail how each of the classifiers was constructed. Along with this descriptive information, full tables with hyperparameters that were tuned can be found in the appendix. Based on the literature review, we have chosen to test Logistic regression, Random forests, K-nearest neighbours, linear discriminant analysis, multilayer perceptron, and Gaussian Naïve Bayes. The selected algorithms are summarised in Table 26.

A graphical representation of the machine learning pipeline can be found in Figure 4.

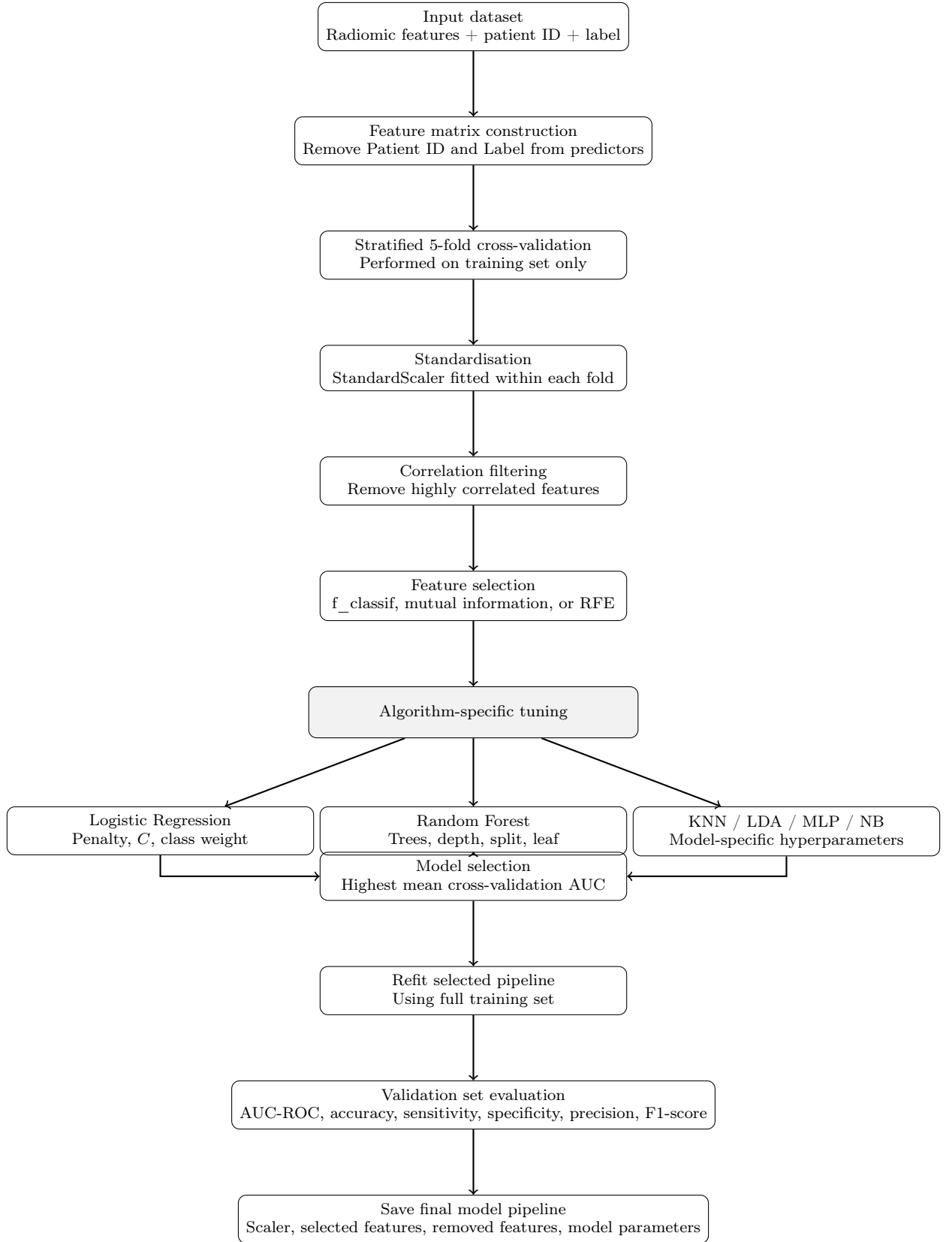


Figure 4: General machine-learning pipeline used for model development. The pre-processing framework was shared across models, while the classifier-specific tuning stage was adapted depending on the algorithm.

0.4.2 Machine Learning Algorithms

0.4.2.1 Logistic regression

Before training, all features were standardised using `StandardScaler` because Logistic Regression is sensitive to feature scale. A correlation filter was then applied to reduce redundant predictors. Several correlation thresholds were tested, and features above each threshold were removed before model fitting. The model was tuned using different regularisation strategies and regularisation strengths. Both LASSO/L1 and Ridge/L2 penalties were tested. LASSO/L1 can perform feature selection by shrinking some coefficients exactly to zero, while Ridge/L2 keeps all features but reduces the size of their coefficients. The regularisation strength was controlled using the `C` parameter, where smaller values apply stronger regularisation and larger values allow the model to fit the data more flexibly. For each combination of correlation threshold, penalty type, and `C` value, the model was trained on the training instances and evaluated on validation instances using AUC. The best model was then selected based on the highest instance-level validation AUC. 5-Fold cross-validation was performed on the Validation Set on instance level.

Table 27 summarises the hyperparameters tested for the Logistic Regression model.

0.4.2.2 Random forest

The Random Forest classifier was built using feature selection and model tuning. Recursive Feature Elimination (RFE) was first applied to select the most informative subset of features. Hyperparameter tuning was then performed using cross-validation on the training set only. Different hyperparameters were tested. A grid search included different numbers of trees, maximum tree depths, minimum samples required to split a node and minimum samples required at each leaf node. The number of trees controls how many decision trees are combined in the forest, while the maximum depth controls how complex each individual tree can become. The minimum split and leaf parameters help control overfitting by limiting how easily trees can create very small branches. The whole grid search took more than one hour.

Table 28 summarises the hyperparameters tested for the Random Forest model.

0.4.2.3 K-Nearest Neighbors

The KNN model was tuned using both preprocessing settings and classifier hyperparameters. Hyperparameter tuning was performed using 5-fold stratified cross-validation on the training set only. Within each fold, standardisation, correlation filtering, feature selection, and model fitting were performed independently to avoid data leakage. Highly correlated features were removed using different thresholds. The remaining features were ranked using either the ANOVA F-test or mutual information. The KNN classifier was tuned over different numbers of neighbours, weighting schemes, and distance metrics. The optimal model was selected based on the highest mean AUC across the five cross-validation folds.

The KNN tuning strategy is shown in Table 29.

0.4.2.4 Linear Discriminant analysis

LDA is affected by feature scale, the features were standardised before training. Model development was performed using 5-fold stratified cross-validation on the training set. A correlation filter was then applied to remove highly correlated features and reduce redundancy. After this, feature selection was performed to test whether a smaller subset of features could maintain or improve classification performance. The model was tuned by testing different numbers of selected features, feature selection methods, LDA solvers, shrinkage settings, and class priors. The number of selected features controlled model complexity. The feature selection method determined how the most informative features were chosen. The solver controlled the mathematical method used to fit LDA. Shrinkage was tested because it can stabilise the covariance estimate, which is useful when radiomic features are numerous and correlated.

The LDA tuning strategy is shown in Table 30.

0.4.2.5 Multilayer perceptron

For the MLP model, a structured training pipeline was constructed using only the training data during model selection. First, the feature matrix was separated from the patient identifier and class label, after which all features were standardised using StandardScaler. Highly correlated features were then removed using different correlation, followed by univariate feature selection using either `f_classif` or `mutual_info`. The number of retained features varied. The MLP architecture was tuned by varying the hidden layer structure, activation function, regularisation strength, and initial learning rate, while the Adam optimiser, adaptive learning rate, early stopping, a maximum of 1000 iterations, and a fixed random state were used. Model selection was performed using stratified 5-fold cross-validation on the training set, with mean AUC used to identify the best configuration. The selected pipeline was then refitted on the full training set. The final trained model was saved together with the scaler, removed correlated features, feature selector, selected features, and model parameters for reproducibility

The LDA tuning strategy is shown in Table 31.

0.4.2.6 Gaussian Naive bayes

For the Gaussian Naïve Bayes model, a structured pipeline was used to tune both preprocessing and classifier-specific parameters. The feature matrix was first created by excluding the patient identifier and class label, and the training data were evaluated using stratified 5-fold cross-validation. Within each fold, features were standardised using StandardScaler, after which highly correlated variables were removed using different correlation thresholds. Feature selection was then applied using SelectKBest, with either `f_classif` or mutual information as the scoring method, and the number of retained features varied. The GaussianNB model was tuned by testing variance smoothing values over a logarithmic range from 10^{-11} to 10^{-1} and by comparing empirical class priors with balanced priors. For each configuration, predicted probabilities were used to calculate the AUC-ROC on each fold, and the final model setting was selected based on the highest mean cross-validation AUC

The GNB tuning strategy is shown in Table 32.

0.4.3 Evaluation Metrics

In order to evaluate the performance of all machine learning algorithms, the following evaluation metrics have been used: accuracy, sensitivity, specificity, precision, F1-score, area under the curve, and receiver operating curve. The final decision about the performance of classifiers has been made based on AUC. A confusion matrix was constructed both on the validation and test sets for instances and on the patient level.

- Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

- Accuracy

Accuracy measures the proportion of all predictions that are correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy gives an overall indication of how often the machine learning algorithm makes the correct prediction. However, it may be misleading when the dataset is imbalanced.

- Sensitivity

Sensitivity, also called recall or the true positive rate, measures the proportion of actual positive cases that are correctly identified.

$$\text{Sensitivity} = \text{Recall} = \text{TPR} = \frac{TP}{TP + FN}$$

Sensitivity shows how well the algorithm detects the positive class. A high sensitivity means the model misses fewer positive cases.

- Specificity

Specificity measures the proportion of actual negative cases that are correctly identified.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Specificity shows how well the algorithm detects the negative class. A high specificity means the model produces fewer false positive predictions.

- Precision

Precision measures the proportion of predicted positive cases that are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Precision shows how reliable the algorithm's positive predictions are. A high precision means that when the model predicts the positive class, it is usually correct.

- F1-score

The F1-score combines precision and sensitivity into one metric. It is the harmonic mean of precision and sensitivity.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}$$

Equivalently:

$$F1 = \frac{2TP}{2TP + FP + FN}$$

The F1-score is useful when there is a need to balance false positives and false negatives. It is often used when the dataset is imbalanced.

- AUC-ROC

The ROC curve plots the true positive rate against the false positive rate at different classification thresholds.

The true positive rate is:

$$TPR = \frac{TP}{TP + FN}$$

The false positive rate is:

$$FPR = \frac{FP}{FP + TN}$$

The area under the ROC curve is:

$$AUC = \int_0^1 TPR(FPR) d(FPR)$$

AUC-ROC measures how well the algorithm separates the positive and negative classes across different decision thresholds. A higher AUC indicates better class separation.

0.5 Results

0.5.1 Model Selection Results

Each model was selected using 5-fold cross-validation on the training set. Hyperparameters and feature-selection settings were tuned to maximise cross-validation AUC while avoiding overfitting. The chosen configurations for every classifier are summarised in the individual model sections below; a direct comparison of the validation-set performance across all models follows in Section 0.5.3.

0.5.2 Individual Model Performance

0.5.2.1 Logistic Regression

The best result with selected hyperparameters is presented in Table 3. This model was, however, not used as the final model due to the number of features selected (122 features, corresponding to 80% of the dataset). To reduce the risk of overfitting and improve generalisability, a more simple model was chosen for evaluation on the validation and test sets.

Table 3: Best Logistic Regression model configuration.

Parameter	Best value
Penalty	LASSO / L1
Correlation limit	0.98
C value	3.03920
Features selected	122
Validation AUC	0.872

The selected Logistic Regression model used LASSO/L1 regularisation with a correlation limit of 0.98 and a regularisation parameter of $C = 0.08532$. After feature selection, 55 features were retained for model training, as summarised in Table 4.

Table 4: Chosen Logistic Regression model configuration.

Parameter	Value
Penalty	LASSO / L1
Correlation limit	0.98
C value	0.08532
Features selected	55
Validation AUC	0.866

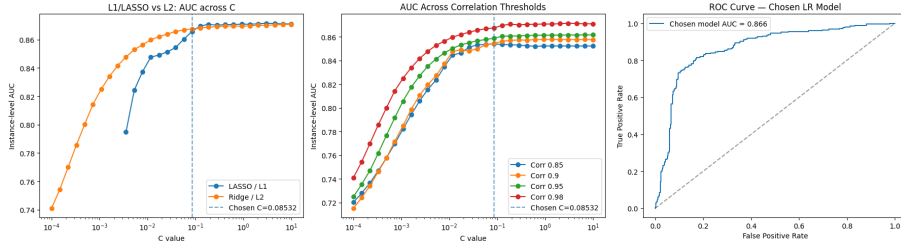


Figure 5: Logistic Regression model selection and evaluation. The left subplot compares L1/LASSO and L2/Ridge regularisation across different C values. The middle subplot shows AUC performance across different correlation thresholds. The right subplot shows the ROC curve for the selected model.

The model achieved a validation AUC of 0.866, indicating good ability to distinguish between benign and malignant cases. Instance-level and patient-level results on the validation set are reported in Tables 5a–6b.

(a) Instance-level validation results for the Logistic Regression model.

Metric	Value
Accuracy	0.755
Sensitivity	0.893
Specificity	0.655
Precision	0.654
F1-score	0.755
AUC-ROC	0.866

(b) Instance-level confusion matrix for the Logistic Regression model.

	Predicted Benign	Predicted Malignant
True Benign	218	115
True Malignant	26	217

Table 5: Instance-level validation performance of the Logistic Regression model.

The confusion matrix shows that the model correctly classified 218 benign and 217 malignant cases, while misclassifying 115 benign cases as malignant and 26 malignant cases as benign. The higher sensitivity than specificity indicates the model is better at identifying malignant cases.

At patient level, the model was evaluated using a decision threshold of 0.5 (Table 6a). Sensitivity remained high at 0.889, while specificity was 0.556, suggesting that some benign patients were incorrectly classified as malignant.

(a) Patient-level validation results for the Logistic Regression model.

Metric	Value
Threshold	0.5
Accuracy	0.722
Sensitivity	0.889
Specificity	0.556
Precision	0.667
F1-score	0.762
AUC-ROC	0.827

(b) Patient-level confusion matrix for the Logistic Regression model.

	Predicted Benign	Predicted Malignant
True Benign	5	4
True Malignant	1	8

Table 6: Patient-level validation performance of the Logistic Regression model.

The model correctly classified 5 benign patients and 8 malignant patients, misclassifying 4 benign patients as malignant and 1 malignant patient as benign. Figure 6 shows the corresponding ROC curve at patient level.

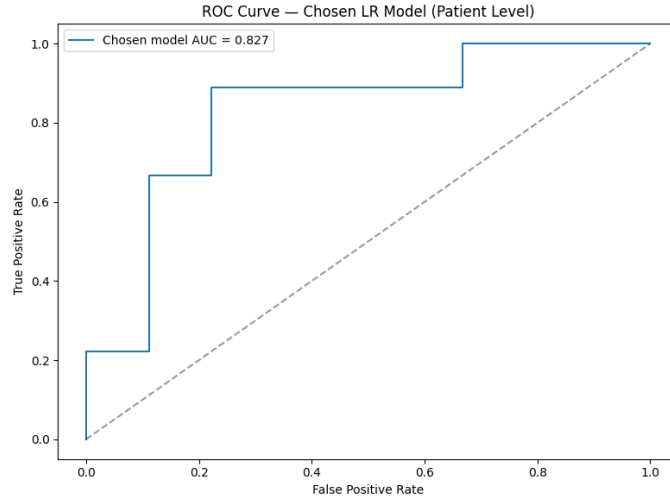


Figure 6: Logistic Regression ROC curve on the validation set at patient level.

0.5.2.2 Random Forest

The Random Forest model was selected using cross-validation on the training set. Figure 9 summarises the selection process, showing how cross-validation AUC varied with the number of RFE-selected features and with different hyperparameter settings. The final configuration is shown in Table 7.

Table 7: Best Random Forest model selected using cross-validation.

Parameter / setting	Selected value
n_estimators	300
max_depth	10
min_samples_split	2
min_samples_leaf	2
Features used	30
Cross-validation AUC	0.843

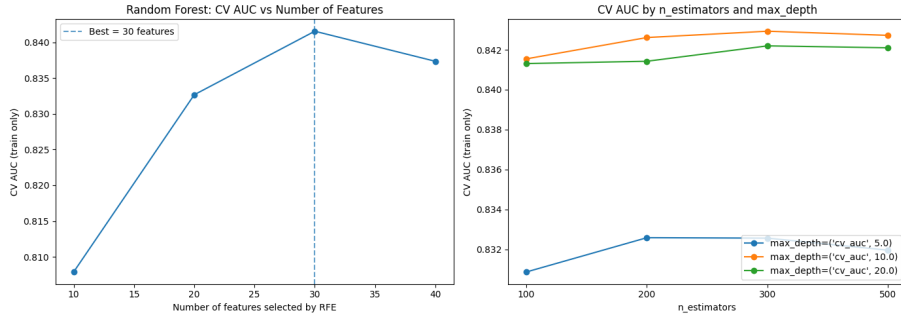


Figure 7: Random Forest hyperparameters tuning results. The left plot shows the effect of the number of RFE-selected features on Cross-validation AUC. The right plot shows AUC for different combinations of n_estimators and max_depth.

The instance-level validation results in Table 8 show strong performance, with an AUC-ROC of 0.894 and accuracy of 0.809. The model correctly classified 261 benign and 205 malignant instances.

(a) Random Forest instance-level validation performance.

Metric	Validation score
Accuracy	0.809
Sensitivity	0.844
Specificity	0.784
Precision	0.740
F1-score	0.788
AUC-ROC	0.894

(b) Random Forest instance-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	261	72
True Malignant	38	205

Table 8: Instance-level validation performance of the Random Forest model.

The model correctly classified 261 benign patients and 205 malignant patients, misclassifying 72 benign patients as malignant and 38 malignant patient as benign. Figure 8 shows the corresponding ROC curve at instance level.

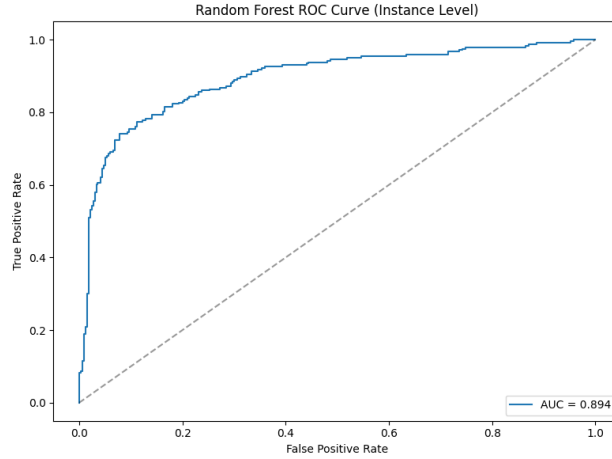


Figure 8: Random Forest hyperparameters AUC results on the validation dataset at the instance level.

At patient level (Table 9), the model achieved an AUC of 0.815 and accuracy of 0.722, correctly classifying 6 benign and 7 malignant patients.

(a) Random Forest patient-level validation performance.

Metric	Validation score
Patient-level AUC	0.815
Applied threshold	0.50
Accuracy	0.722
Sensitivity	0.778
Specificity	0.667
Precision	0.700
F1-score	0.737

(b) Random Forest patient-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	6	3
True Malignant	2	7

Table 9: Patient-level validation performance of the Random Forest model.

The model correctly classified 6 benign patients and 7 malignant patients, misclassifying 3 benign patients as malignant and 2 malignant patient as benign. Figure 9 shows the corresponding ROC curve at the patient level.

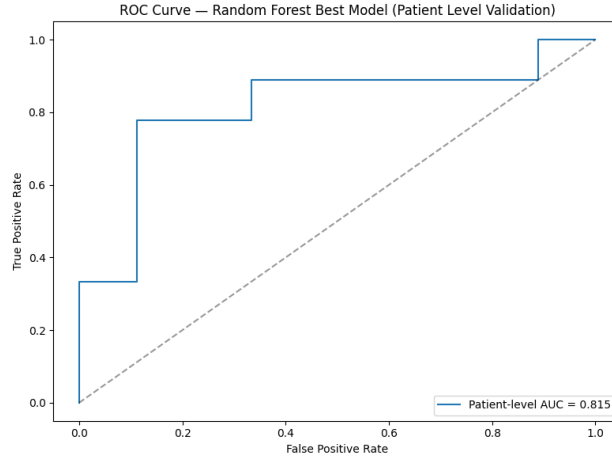


Figure 9: Random Forest hyperparameters AUC results on the validation dataset at the patient level.

0.5.2.3 K-Nearest Neighbours

The KNN model was chosen based on 5-fold cross-validation using mutual information feature selection with distance-weighted Manhattan distance. The selected configuration is summarised in Table 10.

Table 10: Best and chosen KNN model hyperparameters.

Parameter	Selected value
Correlation limit	0.98
Feature selection method	Mutual information
Number of selected features	50
Number of neighbours	30
Weights	Distance
Metric	Manhattan
p	None

Instance-level validation results in Table 11 show an AUC-ROC of 0.871, with 256 benign and 195 malignant instances correctly classified. Figure 10 shows the corresponding ROC curve.

(a) KNN instance-level validation results.

Metric	Value
Accuracy	0.783
Sensitivity	0.802
Specificity	0.769
Precision	0.717
F1-score	0.757
AUC-ROC	0.871

(b) KNN instance-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	256	77
True Malignant	48	195

Table 11: Instance-level validation performance of the KNN model.

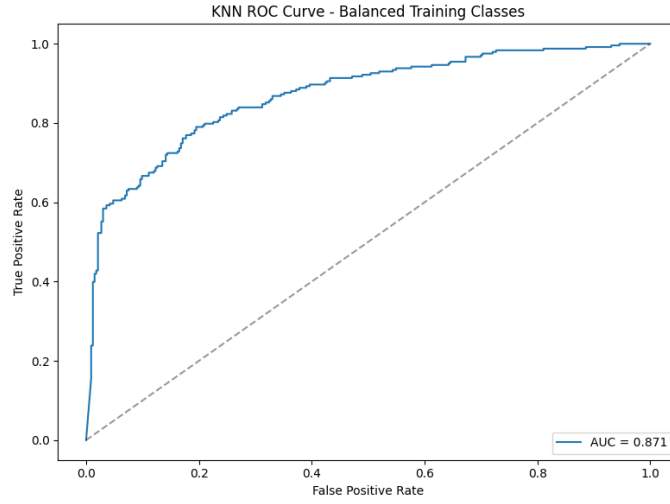


Figure 10: KNN ROC curve on the validation set at instance level.

At patient level (Table 12), the model achieved an AUC-ROC of 0.840 and accuracy of 0.667, correctly classifying 5 benign and 7 malignant patients. Figure 11 shows the patient-level ROC curve.

(a) KNN patient-level validation results.

Metric	Value
Accuracy	0.667
Sensitivity	0.778
Specificity	0.556
Precision	0.636
F1-score	0.700
AUC-ROC	0.840

(b) KNN patient-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	5	4
True Malignant	2	7

Table 12: Patient-level validation performance of the KNN model.

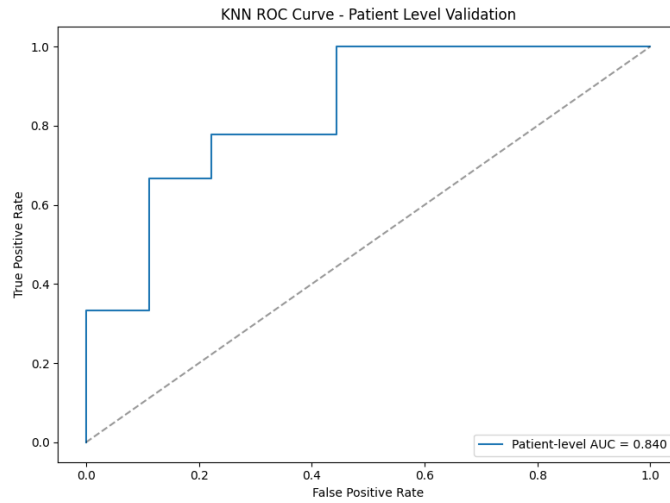


Figure 11: KNN ROC curve on the validation set at patient level.

0.5.2.4 Linear Discriminant Analysis

The best LDA configuration from the initial 5-fold cross-validation search achieved a mean CV AUC of 0.811, using mutual information feature selection with 100 features, LSQR solver, automatic shrinkage, and balanced class priors, as summarised in Table 13.

Table 13: Best LDA model selected using 5-fold cross-validation.

Parameter	Selected value
Correlation threshold	0.95
Feature selection method	Mutual information
Number of selected features	100
Solver	LSQR
Shrinkage	Auto
Class priors	Balanced

However, the final LDA model actually chosen for further evaluation used a correlation threshold of 0.95, 75 selected features, mutual information feature selection, eigen solver, automatic shrinkage, and balanced class priors, as shown in Table 14.

Table 14: Final LDA model actually chosen for evaluation.

Parameter	Selected value
Correlation threshold	0.95
Number of selected features	75
Feature selection method	Mutual information
Solver	Eigen
Shrinkage	Auto
Class priors	Balanced

Instance-level validation results in Table 15 show an AUC-ROC of 0.855, with sensitivity of 0.889 and specificity of 0.667, indicating that the model identified malignant instances more reliably than benign ones.

(a) LDA instance-level validation performance.

Metric	Value
Accuracy	0.760
Sensitivity	0.889
Specificity	0.667
Precision	0.661
F1-score	0.758
AUC-ROC	0.855

(b) LDA instance-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	222	111
True Malignant	27	216

Table 15: Instance-level validation performance of the LDA model.

The model correctly classified 222 benign patients and 216 malignant patients,

misclassifying 111 benign patients as malignant and 27 malignant patient as benign. Figure 12 shows the corresponding ROC curve at instance level.

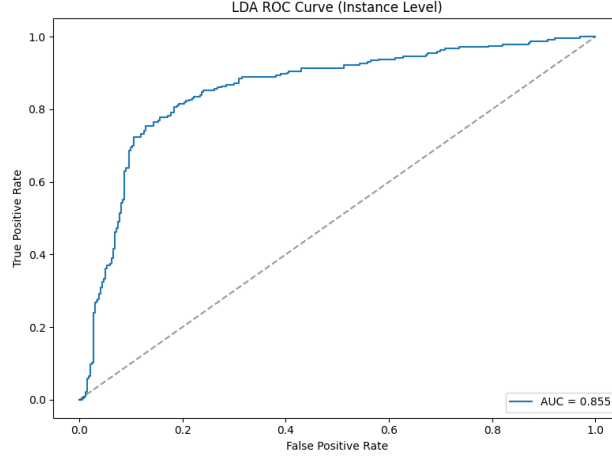


Figure 12: Random Forest hyperparameters AUC results on the validation dataset at the instance level.

At patient level (Table 16), the model achieved an AUC-ROC of 0.790 and accuracy of 0.722, correctly classifying 5 benign and 8 malignant patients.

(a) LDA patient-level validation performance.

Metric	Value
AUC-ROC	0.790
Accuracy	0.722
Sensitivity	0.889
Specificity	0.556
Precision	0.667
F1-score	0.762

(b) LDA patient-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	5	4
True Malignant	1	8

Table 16: Patient-level validation performance of the LDA model.

The model correctly classified 5 benign patients and 8 malignant patients, misclassifying 1 benign patients as malignant and 4 malignant patient as benign. Figure 12 shows the corresponding ROC curve at the patient level.

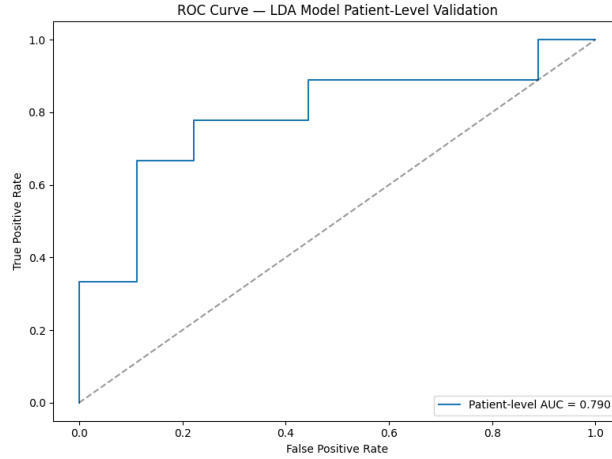


Figure 13: Random Forest hyperparameters AUC results on the validation dataset at the instance level.

0.5.2.5 Multilayer Perceptron

The selected MLP configuration uses moderate feature reduction, two hidden layers, regularisation, adaptive learning, and early stopping to improve generalisation while limiting overfitting. The chosen hyperparameters are listed in Table 17.

Table 17: Best MLP hyperparameters selected using 5-fold cross-validation.

Hyperparameter / setting	Selected value
Correlation limit	0.95
Feature selection method	f_classif
Number of selected features	50
Hidden layers	(64, 32)
Activation	tanh
Solver	adam
Alpha	0.1
Learning rate	adaptive
Initial learning rate	0.01
Early stopping	True

Instance-level validation results in Table 18 show an AUC-ROC of 0.888 and high sensitivity of 0.897, with 227 benign and 218 malignant instances correctly classified.

(a) MLP instance-level validation results.

Metric	Value
Accuracy	0.773
Sensitivity	0.897
Specificity	0.682
Precision	0.673
F1-score	0.769
AUC-ROC	0.888

(b) MLP instance-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	227	106
True Malignant	25	218

Table 18: Instance-level validation performance of the final MLP model.

The model correctly classified 227 benign and 218 malignant instances, while misclassifying 106 malignant and 25 benign instances. Figure 14 shows the patient-level ROC curve.

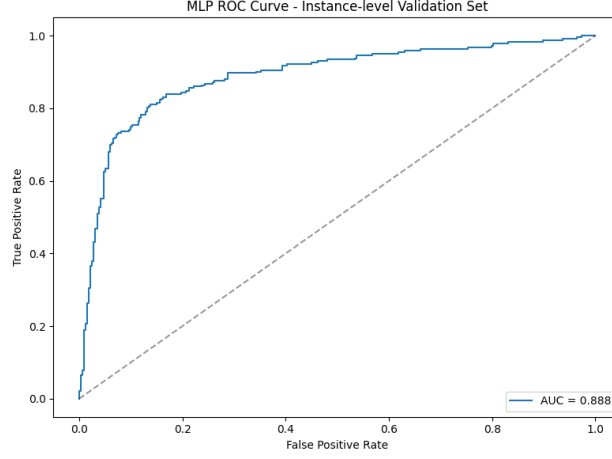


Figure 14: MLP hyperparameters AUC results on the validation dataset at the instance level.

At patient level (Table 19), the model achieved an AUC-ROC of 0.802 and an accuracy of 0.722, correctly classifying 5 benign and 8 malignant patients.

(a) MLP patient-level validation results.

Metric	Value
Accuracy	0.722
Sensitivity	0.889
Specificity	0.556
Precision	0.667
F1-score	0.762
AUC-ROC	0.802

(b) MLP patient-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	5	4
True Malignant	1	8

Table 19: Patient-level validation performance of the final MLP model.

The model correctly classified 5 benign and 8 malignant patients, while misclassifying 4 malignant and 1 benign patients. Figure 15 shows the patient-level ROC curve.

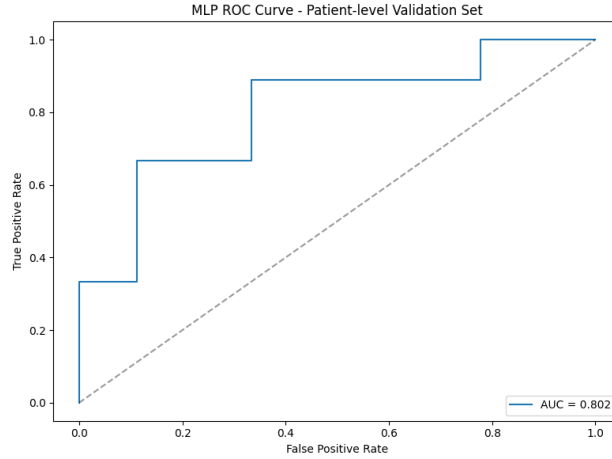


Figure 15: MLP hyperparameters AUC results on the validation dataset at the instance level.

0.5.2.6 Gaussian Naive Bayes

Table 20 compares the selected Gaussian Naive Bayes configuration with the best-performing configuration found during validation. Both models used the same correlation threshold, number of selected features, feature selection method, and prior setting; the main difference was the variance smoothing value.

Table 20: Chosen and best Gaussian Naive Bayes model hyperparameters.

Hyperparameter / setting	Chosen model	Best model
Correlation threshold	0.95	0.95
Number of selected features	25	25
Feature selection method	mutual_info	mutual_info
Variance smoothing	3.593×10^{-6}	4.641589×10^{-5}
Prior setting	Balanced	Balanced

Instance-level validation results in Table 21 show a moderate AUC-ROC of 0.715, with 228 benign and 154 malignant instances correctly classified. A notable number of malignant cases were misclassified as benign. Figure 16 shows the instance-level ROC curve.

(a) Gaussian Naive Bayes instance-level validation performance.

Metric	Value
AUC-ROC	0.715
Accuracy	0.663
Sensitivity	0.634
Specificity	0.685
Precision	0.595
F1-score	0.614

(b) Gaussian Naive Bayes instance-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	228	105
True Malignant	89	154

Table 21: Instance-level validation performance of the Gaussian Naive Bayes model.

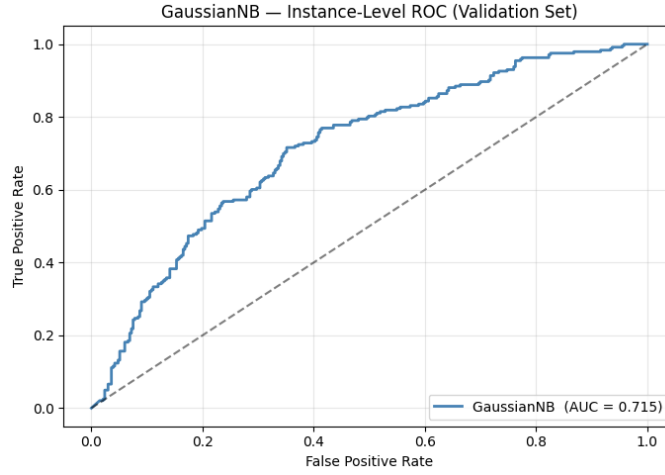


Figure 16: Gaussian Naive Bayes ROC curve on the validation set at instance level.

At patient level (Table 22), performance dropped further, with an AUC-ROC of 0.642 and accuracy of 0.556. The model correctly classified 5 benign and 5 malignant patients while misclassifying 4 patients from each class, indicating near-chance discrimination. Figure 17 shows the patient-level ROC curve.

(a) Gaussian Naive Bayes patient-level validation performance.

Metric	Value
AUC-ROC	0.642
Accuracy	0.556
Sensitivity	0.556
Specificity	0.556
Precision	0.556
F1-score	0.556

(b) Gaussian Naive Bayes patient-level confusion matrix.

	Predicted Benign	Predicted Malignant
True Benign	5	4
True Malignant	4	5

Table 22: Patient-level validation performance of the Gaussian Naive Bayes model.

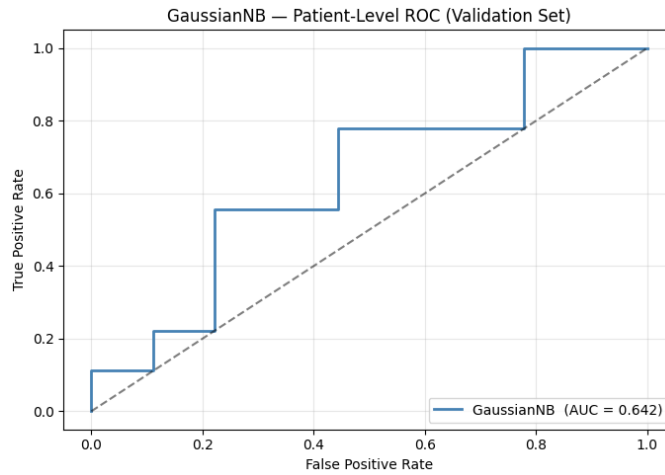


Figure 17: Gaussian Naive Bayes ROC curve on the validation set at patient level.

0.5.3 Comparison of All Models

Table 23 reports performance at the instance level.

Table 23: Instance-level validation results.

Metric	Logistic Regres- sion	Random Forest	KNN	LDA	Naive Bayes	MLP
Accuracy	0.755	0.809	0.783	0.760	0.663	0.773
Sensitivity	0.893	0.844	0.802	0.889	0.634	0.897
Specificity	0.655	0.784	0.769	0.667	0.685	0.682
Precision	0.654	0.740	0.717	0.661	0.595	0.673
F1-score	0.755	0.788	0.757	0.758	0.614	0.769
AUC-ROC	0.866	0.894	0.871	0.855	0.715	0.888

Figure 18 overlays the patient-level ROC curves for all six classifiers on the validation set, providing a direct visual comparison of their discrimination ability after aggregating predictions per patient. Random Forest achieved the highest instance-level AUC-ROC (0.894), while KNN and LDA also demonstrated strong patient-level discrimination (AUC-ROC 0.840 and 0.790 respectively). Gaussian Naive Bayes showed the weakest performance across both evaluation levels. Table 24 presents the corresponding patient-level results after aggregating predictions for each patient.

Table 24: Patient-level validation results.

Metric	Logistic Regres- sion	Random Forest	KNN	LDA	Naive Bayes	MLP
Accuracy	0.722	0.722	0.667	0.722	0.556	0.722
Sensitivity	0.889	0.778	0.778	0.889	0.556	0.889
Specificity	0.556	0.667	0.556	0.556	0.556	0.556
Precision	0.667	0.700	0.636	0.667	0.556	0.667
F1-score	0.762	0.737	0.700	0.762	0.556	0.762
AUC-ROC	0.827	0.815	0.840	0.790	0.642	0.802

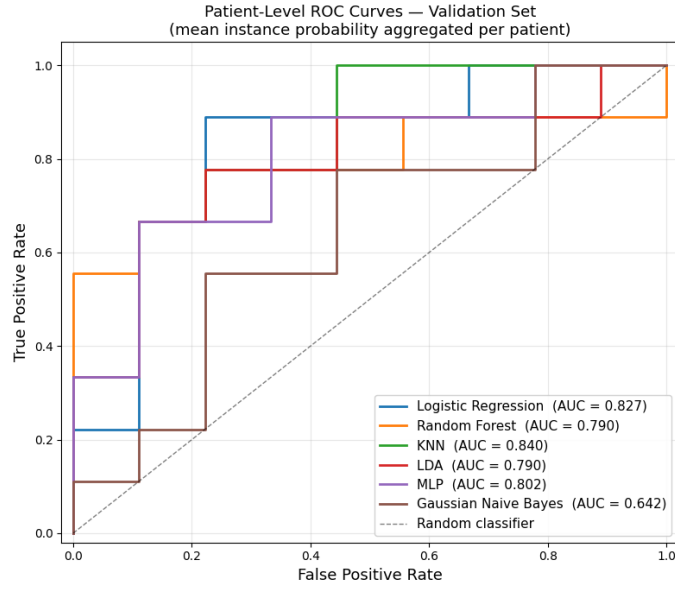


Figure 18: ROC curves for all classifiers on the validation set at patient level.

0.5.4 Final Test Set Results

The final patient-level test set results are summarised in Table 25. Each model was evaluated on 8 patients after aggregating predictions at patient level. Random Forest, KNN, LDA, and MLP achieved the highest AUC-ROC values of 0.750, while LDA showed the highest sensitivity and F1-score. Gaussian Naive Bayes showed the weakest overall test performance with an AUC-ROC of 0.125.

Table 25: Final patient-level test set performance across all classifiers.

Model	AUC-ROC	Acc.	Sens.	Spec.	Prec.	F1	TN	FP	FN	TP
Logistic Regression	0.625	0.625	0.500	0.750	0.667	0.571	3	1	2	2
Random Forest	0.750	0.625	0.500	0.750	0.667	0.571	3	1	2	2
KNN	0.750	0.625	0.500	0.750	0.667	0.571	3	1	2	2
LDA	0.750	0.625	0.750	0.500	0.600	0.667	2	2	1	3
MLP	0.750	0.625	0.500	0.750	0.667	0.571	3	1	2	2
Gaussian Naive Bayes	0.125	0.375	0.250	0.500	0.333	0.286	2	2	3	1

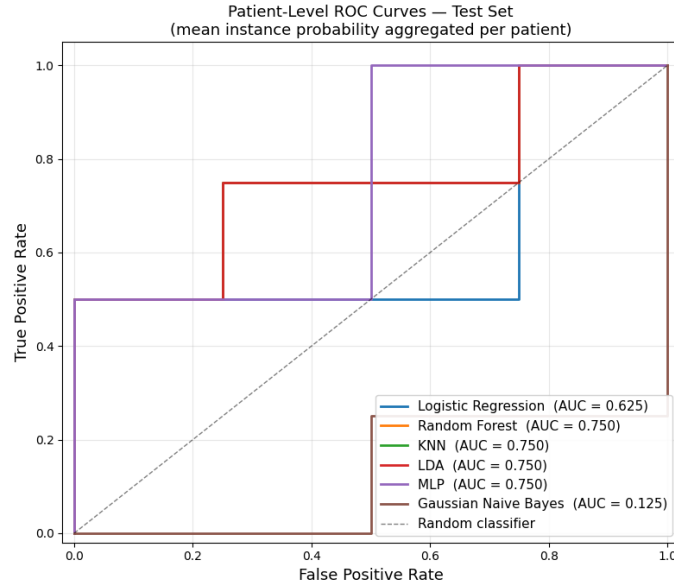


Figure 19: ROC curves for all classifiers on the test set at patient level.

0.5.5 Final Patient-Level Test Set Results

As shown in Table 25, most models achieved comparable patient-level accuracy of 0.625 on the test set. LDA was the only model to favour sensitivity over specificity (sensitivity 0.750, specificity 0.500), correctly identifying 3 out of 4 malignant patients while misclassifying 2 benign patients. Logistic Regression, Random Forest, KNN, and MLP all produced identical test set predictions, each correctly identifying 2 malignant patients and 3 benign patients. Gaussian Naive Bayes performed substantially below all other models, with an AUC-ROC of 0.125 and accuracy of 0.375, indicating near-random and partially inverted discrimination on the test set.

0.6 Discussion

0.6.1 Models discussion

- **Logistic Regression (LR)** The first classifier that was tested was logistic regression it was used because it is easy to interpret and train. The overall impression was that higher results were achieved when highly correlated features were removed. At lower values of parameter c , ridge regularisation outperformed LASSO, while at higher values, there was a slight dominance of Lasso regularisation. In the end, we did not use the best model because it contained 122 features but we chose a different one which used 55 features. Here we can see the correlation of chosen features and labels max positive correlation is around 0.3 and the negative -0.4, in practical terms, this means that features are not really informative and therefore we need to choose more features, but again, they will bring unnecessary complexity and redundancy. LASSO forces the weights of less important features to become exactly zero. The model heavily restricts the size of your feature coefficients.

The Logistic Regression model demonstrated good validation performance at both instance and patient levels. At the instance level, it achieved an accuracy of 0.755 and an AUC-ROC of 0.866, indicating strong overall discrimination between the two classes. Sensitivity was high at 0.893, showing that the model correctly identified most positive instances, although specificity was lower at 0.655, indicating some false-positive classifications. At the patient level, using a threshold of 0.5, the model achieved an accuracy of 0.722 and an AUC-ROC of 0.827. Sensitivity remained high at 0.889, while specificity decreased to 0.556, suggesting that the model was better at detecting positive patients than correctly excluding negative patients. Overall, Logistic Regression appears useful for positive-case detection, but threshold optimisation may be needed to improve specificity and reduce false positives.

- **Random Forest (RF)** Random Forest will randomly sample a subset of these 30 features, model will build 300 independent decision trees with no more than 10 splits from root to leaf and a minimum of 2 data samples at each node. This was not a very simple tree and, therefore forest; however, it should not overfit the data. Trees with 5 or 20 leaves were outperformed, indicating the balance is important. Metrics on the instance based on the validation set: sensitivity was somewhere high which is important; specificity was lower and could be improved; F1-score suggested a reasonable balance between catching positive cases and avoiding too many false positives. The AUC-ROC of 0.894 indicates very good discrimination between the two classes.

On patient base, sensitivity remained relatively high while specificity is lower than sensitivity, meaning the model is more prone to falsely labelling benign patients as malignant than to missing malignant patients. F1 score suggests the model has a fair trade-off between detecting malignant patients and limiting false positives. AUC indicates good overall discrimination between benign and malignant patients but it was lower compared to instance-based classification which makes sense since there is a total of only 18 patients, and we expect metrics to be even lower on test set.

- **K-Nearest Neighbours (KNN)**

The selected number of neighbours was 30, indicating that a relatively smooth decision boundary performed better than a highly local model. Distance weighting was used so that closer neighbours contributed more strongly to the final prediction than more distant neighbours. Manhattan distance was selected as the similarity metric, suggesting that absolute feature differences provided a more effective measure of instance similarity than alternatives such as Euclidean distance. Overall, these hyperparameters indicate that the best KNN model balanced robustness, dimensionality reduction, and local similarity weighting to improve classification performance.

All metrics were relatively high on the instance level, while they were reduced on the patient level. Again, specificity and precision was lower, but AUC was at 0.84

- **Linear Discriminant Analysis (LDA)** In grid search, mutual information appeared to yield better classification results, which in simple terms, means fea-

tures were preferred if they appeared useful for separating the two groups. As the best model not 100 features but rather 75 features were selected. Again, the correlation threshold was at 0.95, it is justifiable that better-performing models had automatic regularisation of the covariance matrix; this is useful when features are numerous, correlated, or when the sample size is not large relative to the number of predictors. It makes the classifier more stable.

On the instance level AUC was pretty high 85% following high sensitivity but on the other hand, low specificity with only 67%, which means that 1/3 of the whole benign dataset was classified as malignant. The F1-score of 0.758 reflects a reasonable balance between sensitivity and precision.

At the patient level, AUC was slightly lower at 79%, sensitivity remained high, while specificity and precision were low. However, specificity was lower at 0.556, showing limited ability to correctly classify negative patients and suggesting a relatively high false-positive rate. Precision was 0.667, meaning that two-thirds of patients predicted as positive were truly positive. The F1-score of 0.762 indicates a reasonable balance between sensitivity and precision. Overall, the model appears better suited to detecting positive patients than to excluding negative patients.

The main goal should be to improve specificity and precision without losing too much sensitivity. This could be done by lowering correlation thresholds, and optimising the number of selected features.

- Multilayer Perceptron (MLP)

Features that were selected using an ANOVA F-test had higher AUC. Those features that differ most strongly between the two classes. This is a supervised feature-selection method suited to classification. A moderate number of 50 features was used, as well as moderate two layer network. The neural network had two hidden layers: the first with 64 neurons and the second with 32 neurons. This gives the model moderate complexity and allows it to learn non-linear patterns. The idea was to try with three layers but due to computational constraints this was not carried out. hyperbolic tangent allowed the network to model non-linear relationships between features and the outcome. Adam is the most commonly used optimiser.

AUC on the instance level is high, reaching 0.88, sensitivity following with 0.89. We can see that specificity and precision were lower. This means around one-third of positive predictions were false positives. The balance between sensitivity and precision was 76.9%, indicating solid positive-class performance.

Overall, the MLP performs well at detecting positive patients, but it is less reliable at identifying negative patients. The high sensitivity of 0.889 suggests the model may be useful when missing positive patients is undesirable. However, the lower specificity of 0.556 indicates that the model produces a relatively high number of false positives.

- Gaussian Naive Bayes (GNB)

The fact that both models selected the same correlation threshold, feature-selection method, number of features, and class-prior setting suggests these

choices were stable during tuning. The main parameter affecting performance was therefore likely variance smoothing. The chosen model applies less variance smoothing, meaning it regularises the feature variance estimates more strongly.

The main reason is probably the assumptions of the model. Gaussian Naive Bayes assumes that features are conditionally independent within each class and that each feature follows an approximately Gaussian distribution which may not be the case. Naive Bayes was chosen as an example of a poorly chosen classifier when data was not examined fully and not understood correctly. However, we want to highlight that we transformations of parameters or for example, using only original data, naive bayes can be tuned to perform well on this dataset but it requires a lot of manual and enginecan transform is around 0.5 which is euqal to random guess which supports our hypothesis that data did not follow a Gaussian distribution and features were selected poorly (might have been depndet which is the case)

0.6.2 Instance level

At the instance level, Random Forest achieved the strongest overall results: it had the highest accuracy at 0.809, highest F1-score at 0.788, highest AUC-ROC at 0.894, highest specificity at 0.784, and highest precision at 0.740. This means it gave the best overall balance between detecting positive cases and correctly identifying negative cases. Although MLP and Logistic Regression had slightly higher sensitivity, Random Forest produced fewer false positives and showed the strongest overall discrimination.

0.6.3 Patient level

At the patient level, the results are more mixed. KNN had the highest AUC-ROC at 0.840, while Logistic Regression, LDA, and MLP had the highest sensitivity and F1-score. However, Random Forest had the best specificity at 0.667 and the best precision at 0.700, suggesting it was better at avoiding false-positive patient classifications

0.6.4 Discussion of Final Patient-Level Test Set Results

On the final patient-level test set, most classifiers achieved modest performance, with Logistic Regression, Random Forest, KNN, and MLP all obtaining an accuracy of 0.625 and specificity of 0.750, but lower sensitivity of 0.500. LDA achieved the highest sensitivity, 0.750, and the highest F1-score, 0.667, indicating better detection of positive patients and the strongest balance between precision and sensitivity. Gaussian Naive Bayes performed poorly, with an AUC-ROC of 0.125 and an F1-score of 0.286, suggesting limited suitability for this classification task. Considering both validation and test performance, LDA would be selected as the preferred model, particularly where detecting positive patients is more important than minimising false positives.

0.6.5 Conclusion

Overall, we are somewhat satisfied with results it took alot of time to train aloghtims chose the best one. We believe our results are not optimal, mainly due to pure feature selection. In some papers, it is said to take all original features as they are ground truth and then add wavelet or LoG features; however, this can induce correlation. We strongly believe that with the right features, all algorithms would have a higher performance. We tried to make trade-offs with the correlation threshold and the number of features. We are aware of purely performance on the test set but we want to see how it performs on unseen data. Also as a final remark, we are aware of the skewnesses of instances per patient which have affected metric evaluation.

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0.7 Appendix

.1 Model Selection

Table 26: Machine learning algorithms used in this project

Algorithm	Algorithm family	Why it is useful for this project
Logistic Regression	Linear model	It is simple, interpretable, and works well as a baseline. It checks whether benign and malignant microcalcifications can be separated using a mostly linear decision boundary.
Random Forest	Tree-based ensemble model	It combines many decision trees and can capture non-linear relationships between radiomic features. It is useful when features interact in complex ways.
Multilayer Perceptron	Neural network model	It learns complex non-linear patterns through hidden layers. It is useful for testing whether deeper feature interactions improve classification.
K-Nearest Neighbours	Instance-based model	It classifies a new microcalcification based on similar examples in the training data. It is useful when samples from the same class are close together in feature space.
Linear Discriminant Analysis	Statistical linear model	It finds a linear projection that maximises class separation. It is interpretable and useful for comparing with other linear methods.
Gaussian Naive Bayes	Probabilistic model	It estimates class probabilities using Bayes' theorem. It is fast, simple, and useful as a probability-based baseline model.

.2 Linear Regression

Table 27: Hyperparameters tested for the Logistic Regression model

Hyperparameter / setting	Values tested	Best value	What it implies
Correlation threshold	0.85, 0.90, 0.95, 0.98	0.90	Controls how strongly correlated two features can be before one is removed. A value of 0.90 means features with correlation above 0.90 were considered redundant.
Regularisation type	L1/LASSO, L2/Ridge	L1/LASSO	Controls how the model handles less useful features. LASSO can shrink some feature coefficients to zero, so it performs feature selection. Ridge keeps all features but reduces their effect.
C value	0.0001 to 10, tested on a logarithmic scale	0.08532	Controls regularisation strength. Smaller C means stronger regularisation. The best value suggests the model needed relatively strong regularisation to reduce overfitting.
Class weight	balanced	balanced	Adjusts the model to handle class imbalance by giving more weight to the minority class.
Solver	liblinear	liblinear	Optimisation method used to train Logistic Regression. It works well for smaller datasets and supports both L1 and L2 regularisation.
Maximum iterations	1000	1000	Sets the maximum number of training iterations. This helps the model converge properly.
Evaluation metric	Validation AUC	0.827	Measures how well the model separates the two classes. An AUC of 0.827 indicates good discriminative performance.

.3 Random forest

Table 28: Hyperparameters and settings tuned for the Random Forest model

Hyperparameter / setting		Values tested	Aim and implication
Feature importance threshold		Mean feature importance	An initial Random Forest with 200 trees was trained on the training set to calculate feature importance scores. Features with importance values greater than or equal to the mean importance were retained, reducing weak or noisy predictors before RFE.
Number of selected features	RFE-	10, 20, 30, 40, 50	Controls how many features are kept after Recursive Feature Elimination. Smaller values reduce model complexity, while larger values preserve more potentially useful predictive information. The best number of features was selected using cross-validation AUC on the training set.
n_estimators		100, 200, 300, 500	Controls the number of decision trees in the forest. More trees can improve prediction stability and reduce variance, but increase computation time.
max_depth		None, 5, 10, 20	Controls the maximum depth of each decision tree. Smaller depths restrict tree complexity and may reduce overfitting, while None allows trees to grow until stopping criteria are met.
min_samples_split		2, 5, 10, 20	Controls the minimum number of samples required to split an internal node. Larger values make the trees less complex and can help reduce overfitting.
min_samples_leaf		1, 2, 4, 8	Controls the minimum number of samples required at a leaf node. Larger values smooth the model by preventing very small terminal nodes, which can improve generalisation.
class_weight		balanced_subsample	Adjusts class weights within each bootstrap sample to reduce the effect of class imbalance during Random Forest training.
Cross-validation		5-fold stratified cross-validation	Used on the training set only for RFE feature selection and hyperparameter tuning. Stratification preserves the class distribution in each fold.
Evaluation metric for tuning		Training AUC cross-validation	Measures how well the model separates benign and malignant instances across classification thresholds. The validation set was not used during tuning.

.4 KNN

Table 29: Hyperparameters and preprocessing settings tuned for the KNN model

Hyperparameter / setting	Values tested	Aim and implication
Correlation threshold	0.85, 0.90, 0.95, 0.98	Removes highly correlated features before training. Lower thresholds remove more redundant features, while higher thresholds keep more information.
Number of selected features	10, 25, 50, 75, 100	Controls how many features are kept after feature selection. Fewer features reduce noise and dimensionality, while more features may preserve more predictive information.
Feature selection method	f_classif, mutual_info	Defines how relevant features are selected. f_classif detects linear class differences, while mutual_info can capture more complex relationships with the label.
n_neighbors	10, 15, 20, 30, 50	Controls how many nearby training instances are used for classification. Smaller values give more local decisions, while larger values produce smoother predictions.
weights	uniform, distance	Controls how neighbours vote. uniform gives all neighbours equal weight, while distance gives closer neighbours stronger influence.
metric	euclidean, manhattan, minkowski	Defines how distance between instances is measured. This is important because KNN classification depends directly on similarity between samples.
p	1, 2	Power parameter for the Minkowski metric. A value of 1 corresponds to Manhattan distance, while a value of 2 corresponds to Euclidean distance.

.5 LDA

Table 30: Hyperparameters and settings tuned for the LDA model

Hyperparameter / setting	Values tested	Aim and implication
Correlation threshold	0.85, 0.90, 0.95, 0.98	Removes highly correlated features before model training. Lower thresholds remove more redundant features, while higher thresholds retain more features.
Number of selected features	10, 25, 50, 75, 100	Controls how many features are kept after feature selection. Fewer features reduce model complexity, while more features may preserve additional predictive information.
Feature selection method	f_classif, mutual_info	Defines how relevant features are selected. f_classif captures linear class-related differences, while mutual_info can capture more complex dependencies between features and the target label.
Solver	svd, lsqr, eigen	Controls the numerical method used to fit the LDA model. Different solvers may behave differently when the feature space is high-dimensional or when features are correlated.
Shrinkage	None, auto, 0.0, 0.1, 0.5, 0.9	Regularises the covariance estimate. In the implementation, shrinkage is only tested with the lsqr and eigen solvers, while the svd solver is used only with no shrinkage.
Priors	empirical, balanced	Controls the assumed class probabilities. Empirical priors use the observed class distribution, while balanced priors use equal class probabilities, corresponding to [0.5, 0.5].
Decision threshold	0.30 to 0.70 in steps of 0.05	Converts predicted probabilities into final class labels. Tuning the threshold allows the model to balance sensitivity, specificity, precision, and recall.
Cross-validation	Stratified 5-fold CV, shuffled, random state = 42	Provides a more robust estimate of model performance while preserving the class distribution in each fold. Shuffling and a fixed random state improve reproducibility.
Evaluation metric	Instance-level validation AUC	Measures how well the model separates benign and malignant instances across thresholds, without relying on one fixed classification threshold.

.6 MLP

.7 Multilayer Perceptron

Table 31: Hyperparameters tested for the MLP model

Hyperparameter / setting	Values tested	What it implies
Correlation threshold	0.90, 0.95	Controls how strongly correlated two features can be before one is removed. Higher thresholds keep more features, while lower thresholds remove more redundancy.
Number of selected features	25, 50, 100	Controls how many features are retained after feature selection. Smaller feature sets reduce dimensionality, while larger sets may preserve more predictive information.
Feature selection method	f_classif, mutual_info	Defines how the most relevant features are selected. f_classif focuses on linear class-related differences, while mutual_info can capture more complex dependencies between features and the target label.
Hidden layer structure	(32), (64), (64, 32)	Defines the number of hidden layers and neurons in each layer. Larger or deeper structures can capture more complex non-linear relationships but may increase overfitting risk.
Activation function	relu, tanh	Determines the non-linear transformation applied in the hidden layers. Different activation functions can affect training stability and how well the model learns complex patterns.
Solver	adam	Optimisation algorithm used to update the network weights. Adam uses adaptive learning rates and is commonly effective for neural networks.
Alpha	0.0001, 0.001, 0.01, 0.1	Controls L2 regularisation strength. Larger values apply stronger regularisation, helping reduce overfitting by penalising large network weights.
Learning rate	adaptive	Adjusts the learning rate during training when progress slows, helping the model continue improving without using a fixed step size throughout training.
Initial learning rate	0.0001, 0.001, 0.01	Sets the starting step size for weight updates. Smaller values train more cautiously, while larger values allow faster updates but may be less stable.
Early stopping	True	Stops training when validation performance no longer improves, reducing the risk of overfitting and unnecessary training iterations.
Cross-validation	Stratified 5-fold CV, shuffled, random state = 42	Preserves the class distribution in each fold and provides a more reliable estimate of model performance.
Evaluation metric	Validation AUC	Measures how well the model separates the two classes across thresholds, without relying on a single fixed classification threshold.

.8 Naive Bayes

Table 32: Hyperparameters tested for the Naive Bayes model

Hyperparameter setting	Values tested	Best value	What it implies
Correlation threshold	0.90, 0.95	0.95	Controls how strongly correlated two features can be before one is removed. A value of 0.95 keeps more features than 0.90 while still removing highly redundant variables.
Number of selected features	25, 50, 75	25	Controls how many features are retained after feature selection. The best value suggests that a smaller feature subset was sufficient and may have reduced noise.
Feature selection method	f_classif, mutual_info	mutual_info	Defines how the most relevant features are selected. f_classif focuses on linear class differences, while mutual_info can capture more complex dependencies between features and the target label.
Variance smoothing	10^{-11} to 10^{-1} , tested on a logarithmic scale with 23 values	4.641589 $\times 10^{-5}$	Adds a small amount of variance to each feature to improve numerical stability. The selected value helps prevent unstable probability estimates when feature variances are very small.
Prior setting	empirical, balanced	balanced	Controls the assumed class probabilities. Empirical priors use the observed class distribution, while balanced priors assign equal probability to both classes.
Cross-validation	Stratified 5-fold CV, shuffled, random state = 42	5-fold stratified CV	Preserves the class distribution in each fold and provides a more reliable estimate of validation performance.
Evaluation metric	Validation AUC	0.715	Measures how well the model separates the two classes across thresholds. The validation AUC indicates moderate discriminative performance.

.9 All Models

Table 33: Performance of selected model configurations on the validation set

(a) Selected model configurations						
Hyperparameter	Logistic gression	Re- est	Random For- est	KNN	LDA	Naive Bayes MLP
Penalty / regularisation	LASSO (L1)	—	—	—	—	Alpha = 0.1
Correlation limit	0.98	—	—	0.98	0.95	0.95
C value	0.08532	—	—	—	—	—
n_estimators	—	300	—	—	—	—
max_depth	—	10	—	—	—	—
min_samples_split	—	2	—	—	—	—
min_samples_leaf	—	2	—	—	—	—
Feature selection method	—	—	mutual_info	mutual_info	mutual_info	f_classif
Features selected / used	55	30	50	75	25	50
Neighbours	—	—	30	—	—	—
Weights	—	—	distance	—	—	—
Distance metric	—	—	manhattan	—	—	—
p	—	—	None	—	—	—
Solver	liblinear	—	—	eigen	—	adam
Shrinkage	—	—	—	auto	—	—
Prior setting	—	—	—	balanced	balanced	—
Variance smoothing	—	—	—	—	3.593×10^{-6}	—
Hidden layers	—	—	—	—	—	(64, 32)
Activation	—	—	—	—	—	tanh
Learning rate	—	—	—	—	—	adaptive
Initial learning rate	—	—	—	—	—	0.01
Early stopping	—	—	—	—	—	True